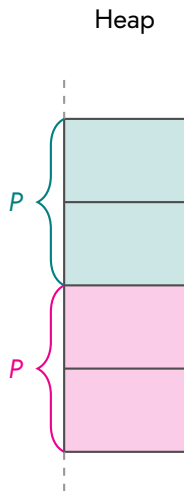


Relational Separation Logic

Heap

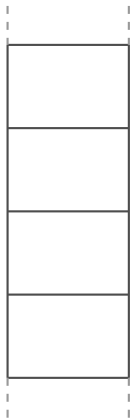


Relational Separation Logic

 $P * Q$ 

Relational Separation Logic

Source Heap

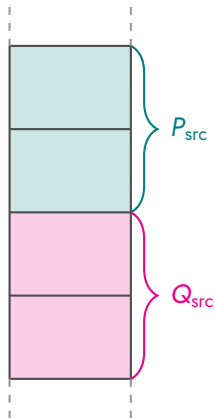


Target Heap



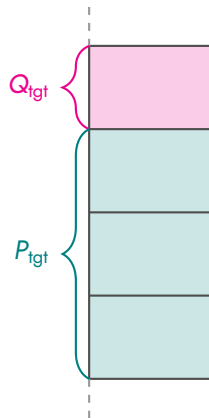
Relational Separation Logic

Source Heap



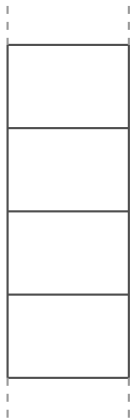
$$P * Q$$

Target Heap

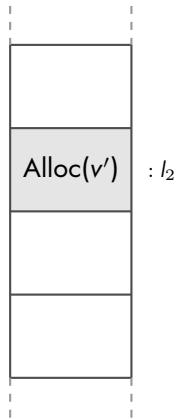


Relational Separation Logic

Source Heap

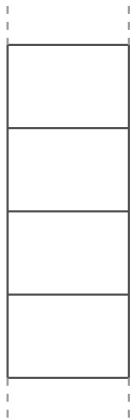


Target Heap



Relational Separation Logic

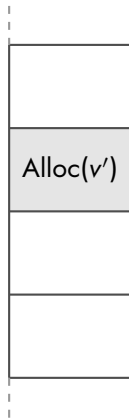
Source Heap



Target Heap

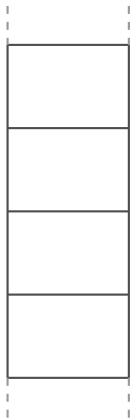
$l_2 \rightarrow_{\mathfrak{t}} v'$

$\text{Alloc}(v') : l_2$



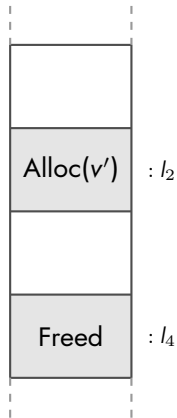
Relational Separation Logic

Source Heap



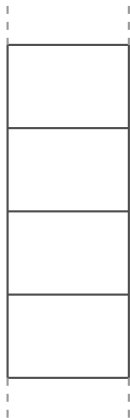
Target Heap

$l_2 \rightarrow_{\dagger} v'$



Relational Separation Logic

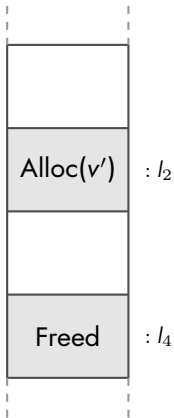
Source Heap



Target Heap

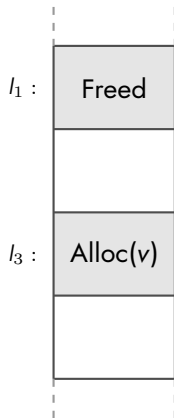
$l_2 \rightarrow_{\mathfrak{t}} v'$

$\text{free}_{\mathfrak{t}}(l_4)$



Relational Separation Logic

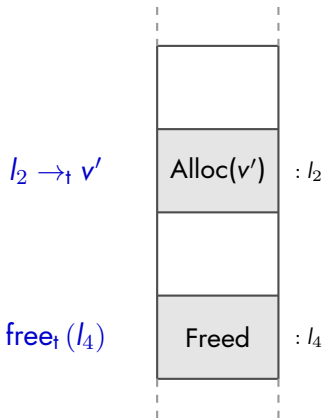
Source Heap



$\text{free}_s(l_1)$

$l_3 \rightarrow_s v$

Target Heap



$l_2 \rightarrow_t v'$

$\text{free}_t(l_4)$